Rocket Engineering Support Proposal

Use this form to propose a project for the rocket club to undertake in support of future rocket development.

Project Title

Custom Flight Computer Design

Project Description

We would design custom flight computer PCBs which would serve as a general go-to for future competition rockets, or as a template for flight computers with additional needs. The goal of this is to open opportunities for making our own flight computers with detailed data-tracking and additional components which aren't accessible through off-the-shelf PCBs. This would use recommendations through thorough research and what past members would find valuable.

This would also create valuable skills for electrical (and possibly software) for future rocketry competitions such as IREC, where precision and electrical trustworthiness is crucial. This project would involve thorough research/planning, designing, digital testing, then manufacturing and physical testing.

My current idea for the design process is where focus is on one PCB with additive stacks with different functions, which is seen in a lot of top collegiate teams. There would be the general stacks that would have common components like the accelerometer, gyroscope, barometer, power, communication, etc., and then the additional stacks could include things like if we wanted to add a camera, temperature sensor, etc. There is a lot of content out there from teams on their flight computer architectures. Other ideas could be where members design different PCBs with specific focuses, but I think creating one main PCB that gets manufactured and then using that as a foundation for additional designs that can just be electrically designed for the time would be best.

[Provide detailed 2-3 paragraphs describing the project. What problem it addresses, what will be built or tested, how it supports development of future rockets]

Estimated Cost and Materials

Research, planning, and design will be very cheap/free, since there are a lot of student resources/licenses available. I would estimate the manufacturing to be roughly $300 with some range. Materials would be PCB components and sensors.

[List estimated total cost and the materials/components needed.]

Timeline

Main goal: Have a fully finished multi-stack PCB manufactured by the end of the school year. Since this isn't on a set competition timeline, it could finish much earlier.

Research and brainstorming ~ 1.5 months

Design and testing ~ 3 months (could be shorter, leaving more room for creating additional designs)

Manufacturing ~ About a week

Testing/additional designs/application (non-school rocket…) ~ Remaining

[Outline the proposed schedule: key milestones, target start/end dates, and how the project fits within the semester.]

Estimated Number of Contributors

As many as interested. There will be plenty of work and even more work can be made to make a more advanced system.

[Estimate how many club members could meaningfully contribute to this project, e.g. 3-5 members]